

Coding of geodesic flow on geometrically finite hyperbolic 3-manifolds with a cusp

1 Coding geodesic flow and limit set

Setup: Let Γ be a geometrically finite subgroup of $\mathrm{SL}_2(\mathbb{R})$ or $\mathrm{SL}_2(\mathbb{C})$ such that $M := \Gamma \backslash \mathbb{H}^d$ (here $d = 2$ or 3) with only one cusp. Suppose $\infty \in \partial \mathbb{H}^d$ is a parabolic fixed point, and let $\Gamma_\infty = \mathrm{Stab}_\infty(\Gamma)$ be the associated parabolic subgroup. We choose Δ_0 a fundamental domain of $\partial \mathbb{H}^d - \{\infty\} \simeq \mathbb{R}^{d-1}$ of Γ_∞ . Let Λ be the limit set of Γ on $\partial \mathbb{H}^d$ and $\Lambda_0 = \Lambda \cap \Delta_0$. Let μ be a Patterson-Sullivan measure supported on Λ . Let δ be the critical exponent.

Recall that we have the Hopf parametrization of the geodesic flow given by $(\partial \mathbb{H}^d \times \partial \mathbb{H}^d - \mathrm{Diag}) \times \mathbb{R} \simeq T^1 \mathbb{H}^d$. With a correct choice of identification, we have the action of Γ is given by

$$\gamma(x, y, t) = (\gamma x, \gamma y, t + \log |D\gamma(x)|), \quad (1.1)$$

where $\log |D\gamma(x)|$ is cohomologous to the usual Busemann function.

We first explain how to obtain a coding of geodesic flow from Proposition 1.1. Here is the main coding result from [LP23] is the following

Proposition 1.1. *There exists a family of disjoint open sets $\{\Delta_j\}_{j \geq 1}$ and an expanding map T on $\cup \Delta_j$ such that*

- $\mu(\Delta_0) = \sum \mu(\Delta_j)$
- For each j , there exists $\gamma_j \in \Gamma$ such that $T|_{\Delta_j} = \gamma_j^{-1}$ and $\gamma_j \Delta_0 = \Delta_j$
- There exists $\lambda < 1$ such that for all j , $|D\gamma_j| \leq \lambda$. (where $|D\gamma_j|$ is the norm of the derivative to the euclidean metric on $\mathbb{R}^{d-1}$)
- (Exponential tail) There exists $\epsilon > 0$ such that

$$\sum_j |D\gamma_j|^{\delta - \epsilon} < \infty.$$

Indeed

$$R(x) = \log |D\gamma_j^{-1}(x)| \quad (1.2)$$

for $x \in \Delta_j$ is the return time and the exponential tail can also be expressed as

$$\int_{\Delta_0} e^{\epsilon R(x)} d\mu(x) < \infty$$

Before going to the coding of flow, we need to add a stable direction to obtain a section of the flow. This is another proposition, which is not hard

Proposition 1.2. *There exists $\Lambda_- \subset \Lambda$ near ∞ such that for all γ_j , we have $\gamma_j^{-1} \Lambda_- \subset \Lambda_-$.*

Fix this identification, we let

$$S = (\cup \Delta_j) \times \Lambda_- \times \{0\}$$

be a section on $T^1\mathbb{H}^d$, which also descends to a section in T^1M . Denote the map by π (Add a graph) **Attention: The coding system we construct later is possibly larger than the geodesic flow. We are not using the first return map to the section.**

We can construct a suspension flow on S by

$$S^R := S \times \mathbb{R}^+ / (x, y, t + R(x)) \sim (Tx, T(x, y), t) \quad (1.3)$$

where for $x \in \Delta_j$, we have $Tx = \gamma_j^{-1}x \in \Delta_0$ and $T(x, y) = \gamma_j^{-1}y \in \Lambda_-$ due to Proposition 1.2. On S^R we have a semiflow on the $\mathbb{R}^+$ direction, denoted by T_t .

Here is the connection between the suspension flow and the geodesic flow (where ν is T -invariant and absolute continuous to the Patterson-Sullivan, comes from general theory of expanding systems or Gibbs measure)

Lemma 1.3. *The map π induces a factor map (S^R, T_t, ν^R) to $(T^1M, \mathcal{G}_t, m^{BMS})$, that is*

$$\pi \circ T_t = \mathcal{G}_t \circ \pi, \quad \pi_*(\nu^R) = m^{BMS}$$

Soft part: The equations Eq. (1.3), Eq. (1.1) and Eq. (1.2) imply that π well defined, since the equivalence relation is mapped to equivalence relation under Γ action, we have

$$\gamma_j(Tx, T(x, y), t) = \gamma_j(\gamma_j^{-1}x, \gamma_j^{-1}y, t) = (x, y, t + \log |D\gamma_j^{-1}(x)|) = (x, y, t + R(x))$$

Since the geodesic flow is mixing, almost every geodesic trajectory will cross this transversal section $\pi(S)$ of T^1M .

Hard part: The proof of equality of measures uses that the T -invariant measure ν on Λ_0 is absolutely continuous to the Patterson-Sullivan measure μ and the product structure, please see [LP23, Prop. 4.15]

Remark 1.4. The delicate thing here is that the map π from S^R to T^1M is not injective, since R may not be the first return time to the section. We only obtain a factor map, but it is still sufficient to conclude the exponential mixing of geodesic flow from suspension flow, see [LP23, Section 4.2]. For the suspension flow, we use a modified Dolgopyat's argument to prove its exponential mixing.

Since we don't need the first return to the section, the advantage is that the construction of coding is more flexible. See also Young tower for a similar situation in the map case.

2 Limit set coding for $SL_2(\mathbb{R})$ and $SL_2(\mathbb{C})$

We start to construct coding of Proposition 1.1. We work only on the boundary $\partial\mathbb{H}^d$. The idea is to use "flowers" near parabolic fixed points. The advantage is that the boundary of flower is clean, which enables us to get good measure estimate to obtain exponential tail.

Let H_0 be the horoball based at infinity. Let $p = \gamma\infty$ be a parabolic fixed point in Δ_0 and $H_p = \gamma H_0$ a horoball based at p . Suppose $\gamma^{-1}\infty \in \Delta_0$. Let h_p be the height of H_p , that is the euclidean height in the upper half space model.

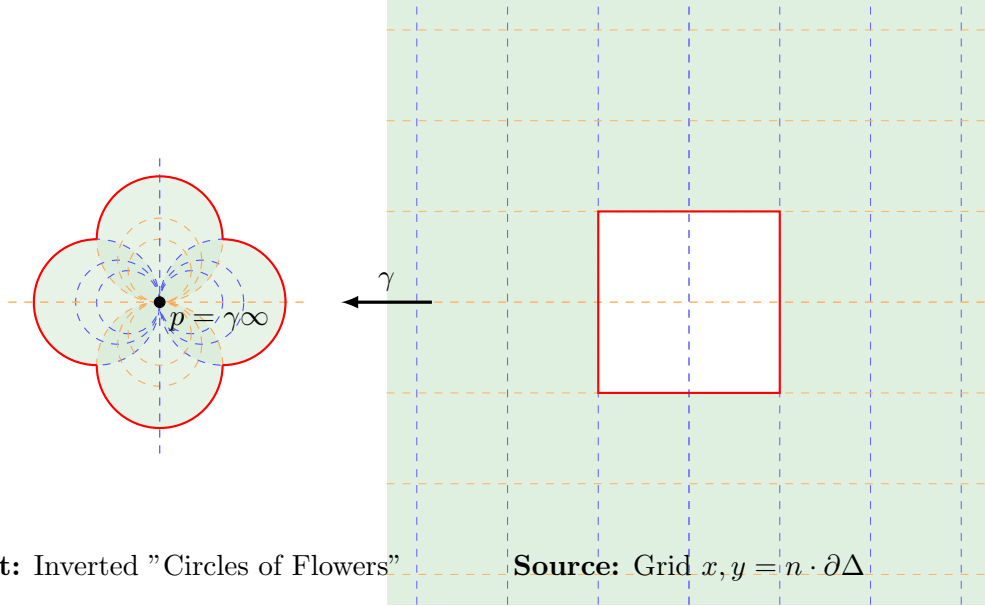
We can tile a neighbourhood of p using a flower J_p : Let $\eta > 0$ be some small parameter. Let F_∞ be the complement of a large parallelotope of size $1/\eta$ tiled by Δ_0 using Γ_∞ . Let

$$N_p = \{\gamma h \in \Gamma_\infty : h \in \Gamma_\infty, h\Delta_0 \notin F_\infty\}, \quad J_p = \gamma F_\infty^c = \bigcup \{\gamma h \Delta_0 : \gamma h \in N_p\}.$$

By computation, we can have

Lemma 2.1. *There exists $C > 0$ such that for any parabolic fixed point $p \in \Delta_0$, we have*

$$B(p, \eta h_p / C) \subset J_p \subset B(p, C \eta h_p)$$



Remark 2.2. For non-maximal rank case, we need modify a bit the definition of J_p . Since the direction without translation action of Γ_∞ is bounded, we take a large set to cover it.

By this way, we find a J_p near a parabolic fixed point and it is a building block of coding. In order to code almost all the set Λ_0 , we need to use flowers of decreasing size. But we cannot be too greedy, we will only take flowers a bit away from the boundary to make sure the remaining set still has good measure estimate.

Here is the way:

- Let $\Omega_0 = \Delta_0$
- For $n \in \mathbb{N}$, let P_n be the set of parabolic fixed points in Ω_n such that $-\log(\eta h_p) \in [n, n+1]$ and $d(p, \partial\Omega_n) > h_p$. (J_p has size e^{-n} , its distance to the boundary is of size e^{-n}/η)
- Let

$$\Omega_{n+1} = \Omega_n - \cup_{p \in P_n} J_p$$

The key proposition is

Proposition 2.3. *There exists $C' > 0$, $\epsilon > 0$ such that*

$$\mu(\Omega_n) \leq C'(1 - \epsilon)^n$$

From this proposition, we can take

$$\{\Delta_j\}_{j \geq 1} = \{\gamma \Delta_0 : \gamma \in N_p, p \in P_n, n \in \mathbb{N}\}$$

3 Mass exchange and exponential tail

This part is devoted to prove Proposition 2.3. We need to go back to the flow and manifold point of view to obtain more information of these flowers.

For a point x in Δ_0 , let $\tilde{x}$ be its lift to $T^1\mathbb{H}^d$ based at ∂H_0 with its forward endpoint equal to x . Let

$$C(\eta) = \bigcup_{p \text{ parabolic fixed}} T^1(H_p(\eta))$$

be the union of horoballs with height ηh_p , whose projection to T^1M is exactly the cusp region.

At time n , the set $\mathcal{G}_n(\tilde{\Omega}_n)$ is a big sheet with many holes coming from "flowers" of different previous steps. For $\mathcal{G}_n(\tilde{x}) \in \mathcal{G}_n(\tilde{\Omega}_n) \cap C(\eta)^c$, we have uniform recurrence of geodesic flow, which enables us to find a new flower near x (Lemma 3.3) in the next fixed steps. For the part $\mathcal{G}_n(\tilde{\Omega}_n) \cap C(\eta)$, we cannot use recurrence. They are the union of the intersection of $\mathcal{G}_n(\tilde{\Omega}_n)$ with different horoballs $H_p(\eta)$. We want to show at next step, a positive proportion of $\mathcal{G}_n(\tilde{\Omega}_n) \cap H_p(\eta)$ goes outside of the cusp $C(\eta)$ (Eq. (3.1)).

This motivates us to introduce the mass exchange argument inspired by Lai-Sang Young.

Definition 3.1. For $n \in \mathbb{N}$

- Let

$$D_n = \cup_{p \in P_n} J_p$$

be part union of flowers at step n .

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$$B_n = \{x \in \Omega_n, \mathcal{G}_n(\tilde{x}) \in C(\eta)\}, \quad A_n = \Omega_n - B_n.$$

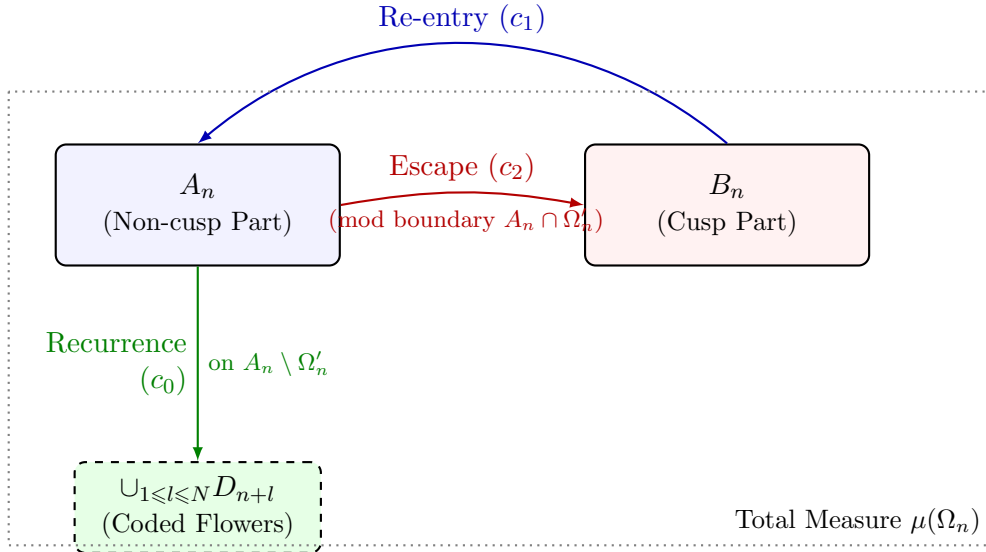
B_n has the same source as flowers, it is the parabolic fixed points too close to the boundary that we cannot take. So they become regions in the cusps.

- We define:

$$\Omega'_n = \{x \in \Omega_n, d(x, \partial\Omega_n) \leq e^{-n}\}$$

the set of points in Ω_n with distance greater than e^{-n} to the boundary.

Mass Exchange Mechanism



Lemma 3.2. • For $n \in \mathbb{N}$, we have

$$B_n \cap D_{n+1} = \emptyset.$$

- (Re-entry part: Mass from cusp to non-cusp region) There exists $c_1 > 0$

$$\mu(B_n \cap A_{n+1}) \geq c_1 \mu(B_n) \quad (3.1)$$

- (Escape part: Mass from non-cusp to cusp region or coded region) There exists $c_2 > 0$ decreasing to 0 as η tending to 0

$$\mu(A_n \cap (D_{n+1} \cup B_{n+1})) \leq c_2 \mu(A_n) + \mu(A_n \cap \Omega'_n) \quad (3.2)$$

Then by the uniform recurrence of the geodesic flow.

Lemma 3.3 (Recurrence). *There exists $N > 0$ and $c_0 > 0$ such that for any $n \in \mathbb{N}$*

$$\mu(\cup_{1 \leq l \leq N} D_{n+l}) > c_0 \mu(A_n - \Omega'_n)$$

We also need a boundary estimate

Lemma 3.4 (Boundary). *There exists $c_3 > 0$ decreasing to 0 as η tending to 0*

$$\mu(A_n \cap \Omega'_n) < c_3 \mu(\Omega_n)$$

We postpone the proofs later and first explain the main mass exchange argument.

Proof of Proposition 2.3. It suffices to show there exists $\epsilon > 0$ such that

$$\mu(\cup_{1 \leq l \leq N} D_{n+l}) > \epsilon \mu(\Omega_n)$$

Step 1: Obtain a uniform bound: Let $t_n = \mu(A_n)/\mu(B_n)$, then by Eqs. (3.1) and (3.2) and Lemma 3.4

$$\begin{aligned} t_{n+1} &= \frac{\mu(A_{n+1})}{\mu(B_{n+1})} = \frac{\mu(A_n) + \mu(B_n \cap A_{n+1}) - \mu(A_n \cap (D_{n+1} \cup B_{n+1}))}{\mu(B_n) - \mu(B_n \cap A_{n+1}) + \mu(A_n \cap B_{n+1})} \\ &\geq \frac{\mu(A_n) + c_1 \mu(B_n) - (c_2 \mu(A_n) + c_3 \mu(\Omega_n))}{\mu(B_n) - c_1 \mu(B_n) + (c_2 \mu(A_n) + c_3 \mu(\Omega_n))} \\ &= f(t_n), \end{aligned}$$

Here f is fractional function which is a convex function. Hence

$$\inf_{t \in \mathbb{R}^+} f(t) \geq \min\left\{\frac{a_1}{b_1}, \frac{a_2}{b_2}\right\} = \min\left\{\frac{1 - (c_2 + c_3)}{c_2 + c_3}, \frac{c_1 - c_3}{1 - (c_1 - c_3)}\right\} = q(c_3).$$

By $t_n > 0$, there is a uniform lower bound of t_n for all $n \in \mathbb{N}$.

Step 2: Then use Lemma 3.3 to obtain the desired statement:

$$\begin{aligned} \mu(\cup_{l=1}^N D_{n+l}) &\geq c_0 \mu(A'_n) \geq c_0 (\mu(A_n) - \mu(A_n \cap \Omega'_n)) \\ &\geq c_0 (\mu(A_n) - c_3 \mu(\Omega_n)) = c_0 \left(\frac{t_n}{1 + t_n} - c_3\right) \mu(\Omega_n). \end{aligned}$$

□

The remaining proofs of lemmas:

Proof of Lemma 3.2. • The set $B_n - B_{n-1}$ is basically the projection of the intersection of $\mathcal{G}_n(\tilde{\Omega}_n)$ with the union of new horoballs H_p with $-\log(\eta h_p) \in [n+1, n+2]$ but the $d(p, \partial\Omega_n) < h_p$. The set $B_{n+1} \cap B_n$ is the union previous bad horoball. By the **separation of horoballs**, we get $B_n \cap D_{n+1} = \emptyset$

- This part comes from $B_n - B_{n+1}$. We separate to disjoint horoballs, and the property comes from regularity of PS measure

$$\mu(B(x, e^{-n/2})) - \mu(B(x, e^{-(n+1)/2})) \geq c_1 \mu(B(x, e^{-n/2}))$$

(But the delicate part is that the connected component in $B_n - B_{n+1}$ may be a portion of an annulus. We introduce equivalent class to deal with this issue.)

- Separation between parabolic fixed points.

□

Proof of Lemma 3.3. We use the uniform recurrence of geodesic flow. For any compact part K in the non-wandering set W , there exists time U such that for any $y \in K$, the unstable leaf $\mathcal{G}_t W^u(y, 1)$ will meet $\partial C(\eta)$ for some $t \in [0, U]$. The proof uses the ergodicity of the geodesic flow.

Then if we take $z = \tilde{x} \in A_n \cap \Lambda - \Omega'_n$ and $y = \mathcal{G}_n(z)$. Then y is in the compact part $W - \Gamma \backslash \Gamma C(\eta)$, the complement of cusp region. By applying the recurrence of $W^u(y, 1)$, we can find a suitable parabolic fixed point p near x . Since x is away from the boundary of Ω_n , this point p will be a good parabolic fixed point and we can take the flower J_p .

(Technical detail: The actual argument is more complicated, since the recurrence only implies $m = \lceil -\log(\eta h_p) \rceil \in [n, n + U]$. When we are at the step m , the point p may not in P_m . But we can analyse the reason why it is not in P_m to find another nearby parabolic fixed point.)

Finally, we can conclude by a doubling property of the measure. $\square$

Proof of Lemma 3.4. Non-concentration and away from boundary, for the expert.

(Skip this part, use directly the soft argument) We use the friendliness of PS measure: There exists $\epsilon, c > 0$, such that for any $r \in [0, 1]$ any l a line in $\mathbb{R}^2$ any $x \in N_{\epsilon r}(l)$, there exists $\rho_x > 0$ such

$$\mu(B(x, \rho_x) \cap (N_r(l) - N_{\epsilon r}(l))) \geq c\mu B(x, \rho_x)$$

Once, we have this estimate, by Vitali covering lemma, we can conclude

$$\mu(N_{\epsilon r}(l)) \leq (1 - c)\mu(N_r(l)).$$

For the geometrically finite case, this friendliness is due to Das-Fishman-Simmons-Urbanski.

(Soft argument) Indeed, we don't need such a stronger version, we only need to bound

$$\mu(A_n \cap \Omega'_n) / \mu(\Omega_n).$$

The points $x \in A_n \cap \Omega'_n$ satisfies $\mathcal{G}_n(\tilde{x})$ in a compact part. Moreover, the radius r in the Ω'_n is around e^{-n} . Hence we are actually dealing with a simpler case.

Here is a soft argument different from the paper: For y in the non-wandering set, let μ_y be the leaf measure on the unstable leaf, which is a scaling of PS measure. We only need to find $1 > \epsilon, c > 0$ such that for any y in a compact region in the non-wandering set and l any line passing through $B(0, \epsilon)$,

$$\mu_y((B(0, 1)) \cap (N_1(l) - N_{\epsilon}(l))) \geq c\mu_y(B(0, 1))$$

This is true due to the fact that PS measure charges no mass on lines, continuity of the fibre measure and the compactness of the region of y .

Then use the geodesic invariance of fiber measure and definition of BMS, we obtain

$$\mu_{\mathcal{G}_n \tilde{x}}(X) = e^{\delta n} (\text{Adg}_n)_* \mu_{\tilde{x}}(X) = e^{\delta n} \mu_{\tilde{x}}(e^{-n} X)$$

for X a measure set in $N^+ \simeq \mathbb{R}^{d-1}$, which implies the desired friendless property for $x \in A_n \cap \Omega'_n$, $\square$

References

- [LP23] Jialun Li and Wenyu Pan. Exponential mixing of geodesic flows for geometrically finite hyperbolic manifolds with cusps. *Inventiones mathematicae*, 231(3):931–1021, March 2023.